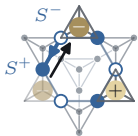


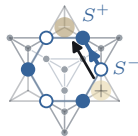
i.a) create


 $S_i^+ S_j^-$ : pair splits

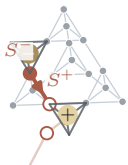
i.b) move


 $S_i^+ S_j^-$ : spinon walks

i.c) annihilate

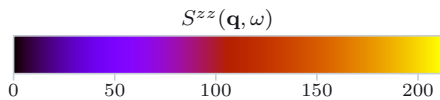

 $S_i^+ S_j^-$ : + meets -  
 $\mathbf{q}_\gamma = (0, 0, 0)$ 

i.d) create

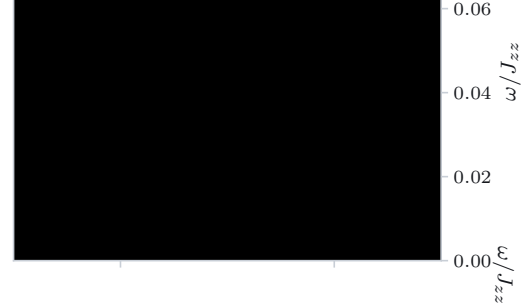

 $S_i^+ S_j^-$ : pair splits
i.e) apply  $S_i^+ S_j^-$ 

the + is carried onward

i.f) identify

PBC closes the open path  
 $\mathbf{q}_\gamma = (0, 0, -1)$ 


iii.a) periodic ED



iii.b) winding-free

